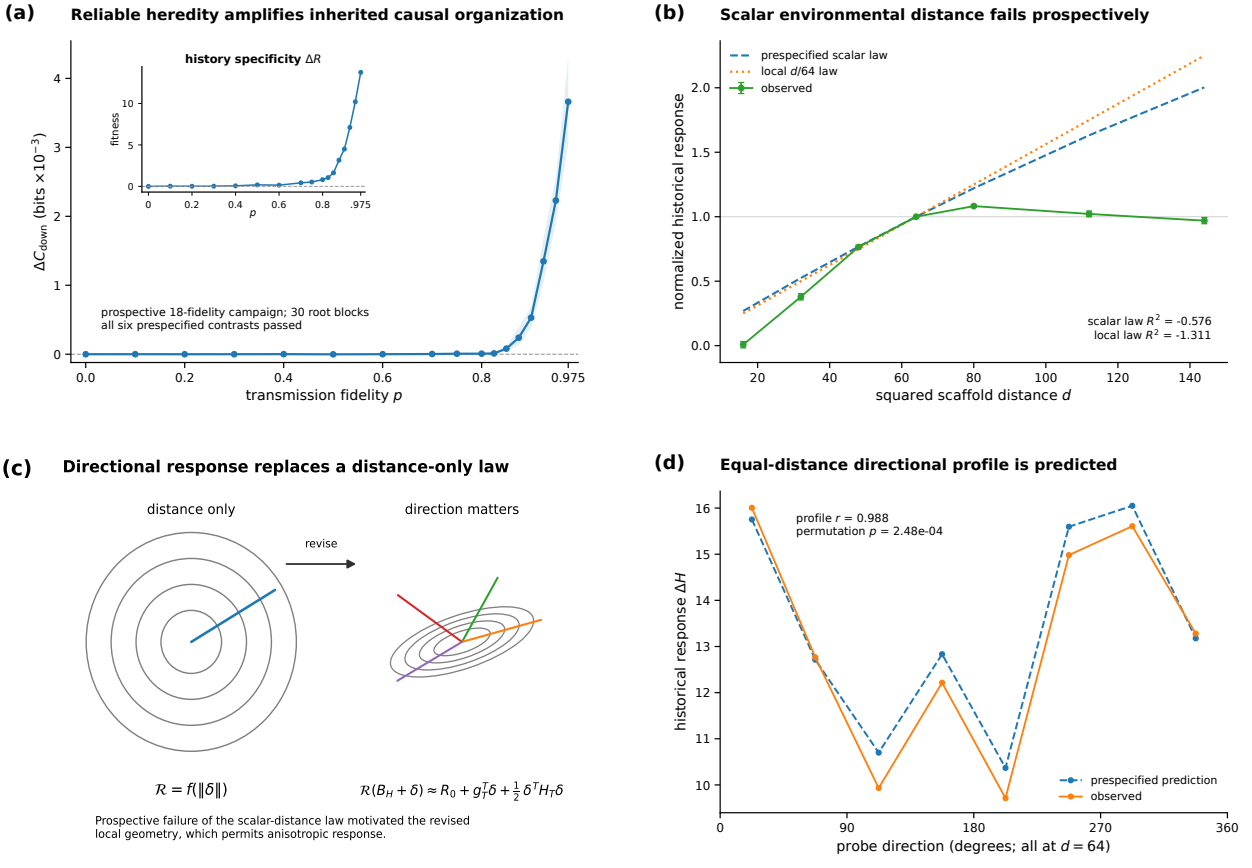

(a)
Reliable heredity amplifies inherited causal organization
history specificity ΔR
fitness
p
ΔC_down (bits ×10^−3)
prospective 18-fidelity campaign; 30 root blocks
all six prespecified contrasts passed
transmission fidelity p
(b)
Scalar environmental distance fails prospectively
prespecified scalar law
local d/64 law
observed
normalized historical response
scalar law R² = -0.576
local law R² = -1.311
squared scaffold distance d
(c)
Directional response replaces a distance-only law
distance only
revise
direction matters
R = f(‖δ‖)
R(B_H + δ) ≈ R_0 + g_T^T δ + ½ δ^T H_T δ
Prospective failure of the scalar-distance law motivated the revised
local geometry, which permits anisotropic response.
(d)
Equal-distance directional profile is predicted
profile r = 0.988
permutation p = 2.48e-04
historical response ΔH
prespecified prediction
observed
probe direction (degrees; all at d = 64)